

Generative Models for Cat Face Synthesis: A Comparative Study of Variational Autoencoders, Generative Adversarial Networks, and Diffusion Models

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Abstract

This paper presents a comparative study of three major generative modeling paradigms – Variational Autoencoders (VAE), Generative Adversarial Networks (GAN), and Diffusion Models – applied to the synthesis of cat face images. We evaluate multiple architectures within each family: β -VAE and VQ-VAE-1 (VAE family), WGAN-GP and SN-GAN (GAN family), and DDIM with a Latent Diffusion Model (diffusion family). The study employs structured grid search over key hyperparameters, training with three fixed random seeds to ensure reproducibility. We evaluate models quantitatively using Fréchet Inception Distance (FID) and qualitatively through specific latent space interpolations and mode collapse analysis. Furthermore, we conduct an exploratory extension training our best model on a mixed dataset of dogs and cats to observe distribution averaging phenomena. This work provides a practical framework for comparing generative paradigms.

Index Terms: generative models, variational autoencoders, generative adversarial networks, diffusion models, image synthesis, cat generation, FID, hyperparameter optimization, mode collapse

1. Introduction

Generative modeling of images has been a central research challenge in machine learning, with three primary paradigms emerging over the past decade: Variational Autoencoders (VAEs) [1], Generative Adversarial Networks (GANs) [2], and Diffusion Models [3]. Each paradigm offers distinct theoretical foundations, training dynamics, and trade-offs in quality, diversity, and training stability. VAEs provide a principled probabilistic framework but often produce blurrier samples; GANs generate sharp images but suffer from training instability and mode collapse; diffusion models have recently demonstrated superior sample quality but require longer inference times.

The selection of appropriate generative models for practical applications remains highly dependent on specific constraints: available computational resources, target sample quality, required inference speed, and stability during training. In this project, we conduct a systematic comparison of representative architectures across all three paradigms on a well-defined task: generating cat face images from the Kaggle Cat Dataset.

Our objectives align with the specific project requirements: (i) standardize varying image resolutions utilizing raw annotation data to accommodate limited computing resources; (ii) systematically compare generation quality across three paradigms (VAE, GAN, Diffusion) using quantitative metrics (FID) and qualitative assessment; (iii) directly address mode collapse through stabilized GAN architectures (WGAN-GP with gradient penalty, SN-GAN with spectral normalization); (iv) analyze

latent space smoothness via interpolation: generating 10-image sequences (2 random endpoints + 8 linear intermediates) to assess semantic continuity; and (v) explore cross-species generalization by training on mixed "Dogs vs. Cats" data to investigate whether the unconditional model learns to separate classes or exhibits distribution averaging (e.g., chimera-like outputs).

2. Methodology

2.1. Datasets and Preprocessing

The primary dataset consists of over 9,000 cat images sourced from the Kaggle Cat Dataset [4]. Raw images exhibit significant variability in resolution, aspect ratio, and face positioning, which poses a challenge for batch processing and limited computing resources. To address this, we apply the following preprocessing pipeline:

1. **Annotation Parsing:** Instead of relying on pre-trained face detectors, we extract the exact 9 coordinate points (eyes, mouth, ears) provided in the corresponding `.cat` annotation files.
2. **Face Cropping:** Bounding boxes are dynamically calculated based on the extreme coordinates of the facial landmarks with an added spatial margin to capture the full face.
3. **Resizing:** Cropped regions are standardized to 128×128 pixels using bilinear interpolation, drastically reducing VRAM footprint and training time.
4. **Normalization:** Pixel values are normalized to the $[0, 1]$ range for VAE/Diffusion models, and to $[-1, 1]$ for GAN models.

Comparison of raw vs. processed images is shown in Figure 1.



Figure 1: Sample images after preprocessing: cropped and resized to 128×128 pixels.

For the exploratory extension, we utilize the Kaggle "Dogs vs. Cats" competition benchmark dataset [5], which contains $\sim$

25,000 mixed images (Figure 2). Since this dataset lacks facial landmark annotations, applying the aforementioned coordinate-based cropping procedure is impossible. Instead, to prevent aspect ratio distortion that could negatively impact the FID metric, we apply a scale-and-crop strategy: resizing the shorter edge of the image to 128 pixels, followed by extracting a 128×128 center crop. This forms a secondary dataset designed to test cross-species distribution averaging.



Figure 2: Sample images from the mixed "Dogs vs. Cats" dataset after scale-and-crop preprocessing.

2.2. Generative Model Architectures

2.2.1. Variational Autoencoders (VAE)

VAEs define a probabilistic generative model via a learned latent variable distribution $q_\phi(z|x)$ parameterized by an encoder network. The generative process is specified by a decoder network $p_\theta(x|z)$ where $z \sim \mathcal{N}(0, I)$. We evaluate two VAE variants:

β -VAE [6]: A standard VAE with convolutional encoder/decoder and a β parameter that controls the trade-off between reconstruction accuracy and prior matching. Grid search varies: latent_dim $\in \{64, 128\}$, $\beta \in \{1.0, 4.0\}$.

VQ-VAE-1: A discrete VAE using vector quantization in the latent space [7]. Instead of a continuous distribution, VQ-VAE discretizes the latent space into a learned codebook to reduce the blurriness typical of standard VAEs. Grid search varies: codebook_size $\in \{512, 1024\}$, feat_map_size $\in \{16 \times 16, 8 \times 8\}$, and reconstruction loss (L1 vs. L2).

2.2.2. Generative Adversarial Networks (GAN)

GANs formulate image generation as an adversarial game between a generator G and discriminator D . To directly address the mode collapse problem required by this study, we avoid the standard DCGAN and evaluate two stabilized variants:

WGAN-GP: Wasserstein GAN [8] with gradient penalty [9] uses the Wasserstein distance and adds a gradient penalty term to enforce the Lipschitz constraint, significantly reducing mode collapse and providing meaningful learning curves. Grid

search varies: n_critic $\in \{3, 5\}$, batch size $\in \{64, 128\}$, learning rate schedule $\in \{\text{symmetric}, \text{TTUR}\}$.

SN-GAN: Spectral Normalization GAN [10] stabilizes discriminator training by normalizing weights with their largest singular value. Grid search varies: batch size $\in \{64, 128\}$, learning rate schedule $\in \{\text{symmetric}, \text{TTUR}\}$, horizontal flip augmentation.

2.2.3. Diffusion Models

Diffusion models define a forward process that corrupts images to noise, then learn to reverse this process via a neural network. To mitigate the immense computational cost of diffusion, we focus on optimized architectures:

DDIM: A faster sampling approach for diffusion models [11] that uses a deterministic sampling procedure, enabling inference in fewer steps. Grid search varies: noise schedule $\in \{\text{linear}, \text{cosine}\}$, base channels $\in \{32, 64\}$.

Latent Diffusion Model (LDM): Following the methodology introduced by Rombach et al. [12], we implement a lightweight diffusion model operating in the latent space of a frozen VQ-VAE encoder. This approach drastically reduces computational overhead while preserving generation quality. Grid search varies sampling steps $\in \{50, 100, 200\}$.

2.3. Training Setup and Hyperparameter Grid

All models employ the Adam optimizer with a default learning rate of $\text{lr} = 2 \times 10^{-4}$ and momentum parameters $(\beta_1, \beta_2) = (0.5, 0.999)$ for stable convergence. Training proceeds for up to 100 epochs with early stopping (patience=15 epochs without improvement on validation loss), enforcing a minimum delta of 10^{-4} for metric improvements. Validation is performed after each epoch using the held-out validation set (10% of the processed cat dataset), and the best model checkpoint is automatically restored after training concludes.

Each hyperparameter configuration is trained independently across three fixed random seeds $\{42, 123, 7\}$ to assess stability and variance across initialization points. Seed management is enforced globally via PyTorch, NumPy, and Python's built-in random module before each training run, ensuring deterministic behavior on the target hardware. Results are subsequently averaged across seeds for robustness assessment.

Table 1: Hyperparameter grid summary across generative model families.

Model Family	Parameter	Search Space
β -VAE	latent_dim, β	$\{64, 128\} \times \{1.0, 4.0\}$
VQ-VAE-1	codebook, map, loss	$\{512, 1024\} \times \{16, 8\} \times \{L1, L2\}$
WGAN-GP	n_critic, batch, LR	$\{3, 5\} \times \{64, 128\} \times \{\text{sym}, \text{TTUR}\}$
SN-GAN	batch, LR, augment	$\{64, 128\} \times \{\text{sym}, \text{TTUR}\} \times \{\text{no}, \text{hflip}\}$
DDIM	schedule, channels	$\{\text{linear}, \text{cosine}\} \times \{32, 64\}$
Latent Diffusion Model (LDM)	steps (inference)	$\{50, 100, 200\}$

2.4. Evaluation Metrics

2.4.1. Fréchet Inception Distance (FID)

FID measures the distance between generated and real image distributions in the feature space of a pre-trained InceptionV3 network. We compute FID on the validation set, averaging results across the 3 seeds for the best configurations.

2.4.2. Latent Interpolation Analysis

To evaluate the smoothness of the learned latent spaces, we execute a strict interpolation protocol: we select two generated images and their corresponding latent noise matrices (z_1 and z_2). We interpolate linearly between these two matrices to generate 8 additional intermediate latent matrices. These 8 matrices are passed through the generator, resulting in a continuous grid of 10 images, which are assessed for semantic continuity.

2.4.3. Exploratory Extension: Mixed Distributions

We train Spectral Normalization GAN (SN-GAN) on a mixed dataset containing both cats and dogs, utilizing a preprocessed dataset of $\sim 25,000$ combined images. The goal is to observe whether the unconditional model learns to separate the classes into mode-specific outputs or suffers from distribution averaging, resulting in generating "chimeras" (e.g., cat faces with dog ears or dog faces with feline features). This experiment probes the limits of unconditional generation under multi-modal input distributions. SN-GAN was deliberately chosen for this task. Diffusion models were too computationally expensive for the $\sim 25,000$ image dataset, and VAEs were rejected because their tendency to blur reconstructions would obscure the distribution averaging effect. SN-GAN provides the sharp, detailed outputs necessary to clearly observe cross-class features in the generated samples.

3. Results

[To be populated]

4. Conclusions

[To be populated]

5. AI Disclosure

This report was edited with AI assistance, to improve clarity, consistency, and formatting. Part of the docstrings, some phrasing and code refactoring or debugging were refined using AI tools, but all scientific decisions and interpretations are the authors' own. The core content, experimental design, and analysis were developed by the authors.

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A. Appendix

A.1. Additional Experimental Details

[To be populated]

A.2. Code Availability

All code, configurations, and training logs are available at: <https://github.com/FlorekPawel/gen-cats>